

IPv6 operations
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N. Buraglio
Energy Sciences Network
O. Caletka
RIPE NCC
J. Linkova
Google
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IPv6-mostly Networks: Deployment and Operations Considerations
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Abstract

This document discusses a deployment scenario called "an IPv6-mostly network", when IPv6-only and IPv4-enabled endpoints coexist ~~on~~ in the same network (~~network segments~~ e.g., local network area (LAN, ~~or~~ VLAN, ~~SSID~~ etc Enterprise network). ~~The~~ This proposed approach enables ~~smooth and~~ incremental transition from dual-stack to IPv6-only networks by allowing IPv6-capable devices to remain IPv6-only while the network is seamlessly supplying IPv4 connectivity to those devices that require it.

Commenté [MB1]: Generalize a bit.

Commenté [MB2]: This is «vague» or «subjective» to assess IMO.

Status of This Memo

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1. Introduction

Most network operators initially deploy IPv6 alongside their existing IPv4 infrastructure, as the dual-stack approach is seen as a necessary transition phase, allowing operators to gain experience with IPv6 while minimizing disruption. Pure IPv6-only networks/connectivity services, where endpoints are not assigned IPv4 addresses and access to IPv4 destinations is provided via some form of address family translation (such as NAT64 [[RFC6146](#)]) remain uncommon outside of the mobile carrier space/operators.

Commenté [MB3]: Not sure what «pure» means here as there are some forms of IPv4 (even in the mobile context as there is transport and user planes).

Commenté [MB4]: Please update to use RFC6146bis

However, dual-stack networks do not address the core problem driving IPv6 adoption: IPv4 address exhaustion. ~~They~~ Such networks still require the same amount of IPv4 address resources as IPv4-only networks. Even worse, ~~this~~ the dual-stack approach has been demonstrated to be a long-term crutch, frequently masking problems that would otherwise be exposed and remedied as normal operational workflows.

The solution to the exhaustion of IPv4 addresses is to stop assigning IPv4 addresses to endpoints, or, in other words, to start deploying IPv6-only networks. To ensure that IPv6-only endpoints can communicate with IPv4-only ones, IPv6-only networks ~~provides~~ provide translation services, such as NAT64 [RFC6146] and SIIT [RFC7755]. However, even with those ~~translation~~ mechanisms in place, many systems and applications retain deep IPv4 dependencies, requiring both IPv4 address assignment to the host ~~and/or~~ the network to provide IPv4 ~~routing functionality~~ forwarding service.

The less control a network operator has over devices and applications, the more difficult it is to eliminate IPv4 dependencies and move to IPv6-only. This is particularly challenging in enterprise networks with legacy IPv4-dependent applications and public Wireless LAN (a.k.a. Wi-Fi) networks where operators cannot guarantee device compatibility. As a result, a chicken-and-egg problem arises: IPv6-only networks seem impractical with so many incompatible applications, yet applications continue to rely on IPv4 because IPv6-only networks are rare.

To enable a gradual transition from dual-stack to IPv6-only, operators need to identify which devices can function in an IPv6-only mode and which cannot. This still leaves the challenge of how to provide IPv4 addresses only to devices ~~which-that~~ can not operate without them. Creating separate network segments for each type of devices introduces complexity and scalability issues - a major hurdle to IPv6-only adoption.

A more desirable approach is to deploy an "IPv6-mostly" network that provides IPv4 on demand only to devices ~~which-that~~ require IPv4 to operate correctly. This allows IPv6-capable devices to remain IPv6-only while the network is seamlessly supplying IPv4 addresses (and connectivity) to those that require it. An IPv6-mostly network allows endpoints to operate at the highest level of their network stack evolution, on demand, while still allowing for legacy IPv4 compatibility.

This document ~~explores~~ defines the requirements, recommendations, and challenges associated with deploying IPv6-mostly networks in enterprise and public Wi-Fi environments. While the principles discussed may be applicable to other network types, this document's focus remains on these specific use cases.

Commenté [MB5]: Other solution are deployed out there with encapsulation using DS-Lite or MAP-E. Please adjust.

Commenté [MB6]: As this claim may not be true for not all network legs.

Commenté [MB7]: It is even challenging in the mass/residential market.

I would delete «particularly».

Commenté [MB8]: -- Suggest to add this to the abstract

2. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

3. Terminology

This document reuses ~~most of terms defined in Section 1.2 of Terminology Section from of~~ [RFC8925].~~→~~

In addition, the document makes use of the following terms:

IPv6-mostly network: A network segment on which IPv6-only and IPv4-enabled endpoints coexist, and where the network provides IPv4 connectivity on demand (e.g., via DHCPv4 Option 108, [RFC8925]) only to the endpoints that require it, while IPv6-capable endpoints operate in IPv6-only mode, using NAT64 ([RFC6146]) or 464XLAT ([RFC6877]) to reach IPv4-only destinations.

CLAT: a customer-side translator (CLAT) as specified in [RFC6877] ~~and [I-D.ietf-v6ops-clatcn]~~.

Endpoint: A device connected to a network and considered a host from the operator's perspective. Examples include:

- * Personal computers: While primarily a host, such devices might run virtual systems/containers, and route traffic to them, extending the network and acting as routers.
- * Mobile phone with tethering enabled: Acts as a host on the Wi-Fi network, but also as a router for tethered devices, potentially without the operator's knowledge or consent.

~~NAT44~~: Network Address Translation from IPv4 to IPv4.

Native IPv4 connectivity: IPv4 connectivity ~~or default gateway~~ provided by the network without using any form of translation mechanisms (such as 464XLAT, [RFC6877]).

Network segment: a link (~~VLAN~~, a broadcast domain, etc.) where hosts share the same IP subnet.

~~PREF64+-(~~ (or NAT64 prefix): An IPv6 prefix used for IPv6 address Synthesis [RFC6052] and for network addresses and protocols translation from IPv6 clients to IPv4 destinations~~→~~ [RFC6146].

4. Solution Overview

In a nutshell, an IPv6-mostly network is very similar to a dual-stack one with two additional key elements:

- * The network provides NAT64 ~~+[RFC6146]+~~ functionality, enabling IPv6-only clients to communicate with IPv4-only destinations.

Commenté [MB9]: The term is already defined in 8925. why are we re-defining it here?

RFC8925 has the following:

«A network that provides NAT64 (possibly with DNS64) service as well as IPv4 connectivity and allows the coexistence of IPv6-only, dual-stack, and IPv4-only hosts on the same segment. Such a deployment scenario allows operators to incrementally turn off IPv4 on end hosts, while still providing IPv4 to devices that require IPv4 to operate. But IPv6-only-capable devices need not be assigned IPv4 addresses.»

Commenté [MB10]: I don't think this is needed for the terminology.

Commenté [MB11]: Consider adding an informative reference to RFC 3022.

Commenté [MB12]: I don't parse this.

Commenté [MB13]: Please expand

- The network also provides the information about the NAT64 prefix (PREF64), for example, via ~~RAS~~ ~~+~~[RFC8781]~~+~~, via DNS64 ([RFC6147], [RFC7050]), or both. This is to ensure that clients and the network's NAT64 use the same PREF64 to translate between IPv6 and IPv4. ~~Section~~~~Sections~~ 4.3.3 and ~~Section~~ 4.3.4 discuss those mechanisms in more detail.

Commenté [MB14]: Please expand

* The DHCPv4 server infrastructure offers IPv6-Only Preferred Option (Option 108) as per [RFC8925]. This is to ensure that IPv6-only-~~capable~~~~capable~~ devices are not consuming IPv4 addresses. Section 4.2 discusses other approaches to provide IPv4 addresses on demand.

Commenté [MB15]: «purist» as there is no such a thing.

If you want to use it, then add an entry to the terminology to say that DHCPv4 is used to refer to DHCP [RFC2131]

Upon connecting to an IPv6-mostly network~~segment~~, an endpoint configures its IP stack based on its capabilities:

* IPv4-only ~~Endpoint~~~~endpoint~~: Acquires an IPv4 address through DHCPv4.

* Dual-~~Stack-stack~~ ~~Endpoint-endpoint~~ (Not IPv6-~~only-only-capable~~~~capable~~): Configures IPv6 addresses using any supported protocol. Additionally, it obtains an IPv4 address via DHCPv4.

* ~~IPv6-only-only-capable-capable~~ endpoint configures its IPv6 addresses and, while performing DHCPv4, includes Option 108 ~~+~~[RFC8925]~~+~~ into the Parameter Request List. ~~If the request is accepted, The-the~~ DHCPv4 server returns the option and, as per [RFC8925]~~+~~, the endpoint forgoes requesting an IPv4 address, remaining in IPv6-only mode.

Commenté [MB16]: For consistency with how this is used and defined in RFC8925

An IPv6-mostly network segment can support a mix of IPv4-only, dual-stack, and IPv6-only ~~endpoints~~~~devices~~. IPv6-only endpoints utilize ~~thea~~ network-provided NAT64 to reach IPv4-only destinations.

Commenté [MB17]: As this is the term you defined in the terminology section.

Commenté [MB18]: As there might be multiple NAT64 instances.

The following sections ~~discussed~~~~discuss~~ those solution elements in more details.

4.1. IPv6-only~~capable~~~~capable~~ Endpoints

~~The term "IPv6-only-only-capable endpoint" lacks a strict technical definition. It broadly describes an IPv6-capable device that can function without native IPv4 connectivity/IPv4 addresses, providing the same user experience as if native IPv4 connectivity were present. Examples include but are not limited to:~~

Commenté [MB19]: -- Do we need this given that a definition is provided in 8925?

Do we lose something if we delete 4.1?

* An endpoint capable of communicating only with a limited set of endpoints, all of which are available over IPv6.

* An endpoint capable of utilizing only IPv6 sockets, employing NAT64 for accessing IPv4 resources.

* An endpoint implementing Customer-side Translator (CLAT) to provide IPv4 connectivity for the legacy applications.

4.2. IPv6-only and IPv4-enabled Endpoints Coexistence

One effective way to restrict IPv4 addresses solely to devices that require them is to enable support for the IPv6-only Preferred Option for DHCPv4 (Option 108, [RFC8925]) on the network's DHCP infrastructure. As most IPv6-only capable devices supports Option 108 in their DHCPv4 clients, enabling Option 108 on the server side would allow those devices to cease requesting IPv4 addresses.

Certain devices, such as resource-constrained embedded systems, may operate in IPv6-only mode without specific adjustments if their communications ~~is~~are limited to IPv6-enabled destinations. However, ~~Since thesesuch~~ systems ~~often may~~ lack Option 108 support. As such, administrators may need alternative methods to prevent IPv4 address assignment. One approach is to block IPv4 traffic at the network attachment level (such as a switchport or wireless access point). This could involve methods like:

* Static ACL: Applying a static filter with, e.g., a "deny ip any any" rule.

* Dynamic ACL via Remote Authentication Dial In User Service (RADIUS): If 802.1x authentication is in use, RADIUS [RFC3580] can provide an ACL blocking all IPv4 traffic.

* Filtering the IPv4 ethertype (0x0800): Some hardware platforms may allow for low level filtering of ethertype 0x0800 preventing IPv4 from passing into or out of a given switchport.

The ACL-based approach presents significant scalability and operational complexity challenges, particularly in environments with highly mobile endpoints. These challenges arise from the requirement that operators define and maintain the specific intent regarding which devices should have IPv4 traffic blocked. Consequently, ~~this the~~ ACL-based approach is recommended only as an interim or stopgap solution.

4.3. Access to IPv4-only Destinations

4.3.1. NAT64 Deployment

IPv6-only endpoints require NAT64 to access IPv4-only destinations. NAT64 function~~ality~~ can be deployed on dedicated devices or by enabling it on devices already performing NAT44. Consolidating NAT44 and NAT64 ~~functionality functions~~ on the same set of devices offers significant

reductions in the cost and complexity of an IPv6-mostly deployment.

If an existing NAT44 equipment supports NAT64 function, the deployment of dedicated NAT64 devices warrants careful consideration of various factors, including, but not limited to:

Commenté [MB20]: -- Do we have a public record of this claim?

If not, I suggest to reword, e.g., «Assuming IPv6-capable devices support...»

Commenté [MB21]: As this may not age well.

Commenté [MB22]: Please expand

Commenté [MB23]: Add a brief text to explain what this rule is meant for.

The reason is that there is no standard way to express ACLs/ACEs/

Commenté [MB24]: Add an informative reference.

Commenté [MB25]: As this depends on the scale of attached devices

Commenté [MB26]: Why do we need to make a reco here?

s/recommended/can be used?

Commenté [MB27]: Some of these considerations are discussed in <https://datatracker.ietf.org/doc/html/rfc7269#section-3.1.4>

Commenté [MB28]: Maybe say somewhere that these don't support IPv4-in-IPv6 encap.

Commenté [MB29]: For consistency as «NAT64 function» is used in the same para

Commenté [MB30]: Let's not speculate about this.

Commenté [MB31]: As it may not be present in a deployment

- * Deployment costs: Purchasing and installing additional devices, including the resulting increased requirements for rack space and power.
- * Network architecture impact: Network design and topology changes necessary to accommodate new devices.
- * Management Integration: Integration of NAT64 device support within network management systems and related automation platforms.
- * **Capacity planning complexity:** When deploying separate devices for NAT64, the operator needs to forecast how many sessions will move from NAT44 to NAT64 during the IPv6-mostly migration. The operator needs also predict how many sessions each type of NAT will handle in the near future. If the NAT64 and NAT44 devices are combined, the NAT devices only need to be scaled to the total number of sessions going to IPv4-only destinations.
- * Operational costs: Ongoing operational costs associated with the new devices.

Commenté [MB32]: This is the key one.

The **Network-network** architecture also influences the decision to deploy dedicated devices. NAT44, for instance, is typically implemented at the network perimeter (Internet edge). However, if internal services are not fully IPv6-enabled, NAT64 is required for IPv6-only **clients endpoints** to access both internal and external IPv4 destinations. In ~~that~~ **such a** case, it might be beneficial to deploy NAT64 **functions** closer to the **users endpoints**, instead of the network Internet edge, in order to simplify routing **and/** firewall rules and reduce latency.

4.3.1.1. NAT64 IPv4 **Address Pools** Considerations

If NAT64 and NAT44 are performed by the same devices, **an operator** must choose between allocating a dedicated IPv4 **address pool** for NAT64 or using the existing NAT44 **address pool** for both. In general, using the same **address pool** for both translation types has a number of advantages:

- * Simplified deployment: no need to provision new **address pools**.
- * No additional public IPv4 **address** space required.
- * Simplified capacity planning. The size of **IPv4 NAT address pool** is scaled to the total number of expected sessions, meaning **that** the operator does not need to forecast (or know the current ratio of) how many sessions will use NAT44 and how many will use NAT64, nor predict future changes in those numbers.

- * **Minimised-Minimized** user impact: it is not uncommon for client IP addresses

Commenté [MB33]: -- This depends on the implementations.

Some offer this using different service cards and this the external IP address may be configured explicitly to each nat instance.

Do we have an implem example where the same pool is used by these instances?

Commenté [MB34]: I guess we mean NAT44+NAT64?

Please make that clear.

Commenté [MB35]: Usage consistency in the doc

to be used as a part of ~~an~~ authentication, identification, and ~~access-control lists~~ (ACLs) on the server side. If the NAT44 address pool is already allowed by an ACL on a remote IPv4-only destination, then using a dedicated NAT64 address pool would change the client's public IPv4 address (the one the destination sees) when the client becomes IPv6-only and starts using NAT64. This change might impact the client's connectivity to that destination until the ACLs on the remote side are updated.

Additionally, web servers are known to use client IP addresses for various forms of authentication, identification and session tracking. Therefore, the

NAT64 SHOULD be configured so, whenever possible, the same NAT64 external IPv4 address pool address is assigned to all concurrent sessions originating from the same IPv6 address of an internal host (a configuration often referred to as 'sticky' NAT). REQ-2 of [RFC4787] and [RFC6888] discusses this behaviour in more details.

4.3.1.2. NAT64 Prefix Considerations

Operators can use either the Well-Known NAT64 Prefix (64:ff9b::/96, WKP) [RFC6052], or they can choose a Network-Specific Prefix (NSP), either from their own address space or from the Local-Use IPv4/IPv6 Translation Prefix (64:ff9b:1::/48, LUP, [RFC8215]). Using the WKP or LUP can make it simpler to set up packet filtering rules and troubleshoot the network, since traffic between IPv6-only hosts and IPv4-only destinations is easy to spot and tell apart from regular IPv6 traffic. Another benefit of the WKP or LUP is that if the client received both non-synthesized (native) and synthesized AAAA responses from DNS, the Rule 9 of the destination address selection algorithm (section 6 of [RFC6724]) would prefer the native IPv6 destination address over a synthesized one. Additionally, using WKP lets the operator use public DNS64 services, which rely on the WKP for address translation.

~~Currently~~ Section 3.1 of [RFC6052] forbids IPv4/IPv6 translators from using the WKP for private [RFC1918] and shared [RFC6598] IPv4 addresses. [I-D.ietf-v6ops-nat64-wkp-1918] discusses the NAT64 prefix selection considerations and ~~proposes-specifies~~ updates to Section 3.1 of [RFC6052], so translators are allowed to use the WKP for those types of non-global IPv4 addresses.

If internal IPv4-only destinations use [RFC1918] or [RFC6598] IPv4 addresses and IPv6-only devices need to reach them, the operator should ensure that their NAT64 ~~functions devices~~ are compliant with the changes proposed in [I-D.ietf-v6ops-nat64-wkp-1918]. Alternatively, the operator might utilize a network-specific NAT64 prefix.

4.3.2. 464XLAT

Unless the usage of IPv4-only network sockets is prohibited or

Commenté [MB36]: Identification based on IP address is a common practice but this is not considered as authentication.

I would delete this as this can distract.

Commenté [MB37]: Modulo the issues in <https://www.rfc-editor.org/info/rfc6269/#section-13.1>

Commenté [MB38]: -- Existing impls already allow to relax that. I don't think we need to cite I-D.ietf-v6ops-nat64-wkp-1918 in particular here.

I would not tie this behavior to this specific draft.

mitigated (e.g., by implementing translation mechanisms such as BIH [RFC6535]) in the ~~operating~~ Operating system (OS), enabling CLAT ~~(Customer-side Translator)~~ on endpoints is essential for seamless operation of IPv4-only applications in IPv6-only environments. CLAT provides an IPv4 address and IPv4 default route, ensuring functionality even without assignment of native IPv4 address ~~from~~ by the network. Without CLAT or similar compatibility layer inside the ~~operating system~~ OS, IPv4-only applications would fail, negatively impacting user experience and increasing support overhead.

Recommendations for ~~n~~Network ~~a~~Administrators controlling the endpoints:

- * CLAT + DHCPv4 Option 108: If the network administrator can control endpoint configuration, CLAT SHOULD be enabled on endpoints sending DHCPv4 Option 108. This streamlines the transition.
- * Option 108 without CLAT MAY be enabled if the administrator aims to identify IPv4-only systems/applications, or if all applications are confirmed to work correctly in IPv6-only mode.

Commenté [MB39]: We use both «network administrator» and «operator» in the document. Maybe clarify how both relate to each other.

Commenté [MB40]: What does that concretely mean?

4.3.3. Signalling NAT64 Prefix to Hosts

Hosts running 464XLAT, translating IPv4 address literals to ~~an~~ IPv6 literals (Section 7.1 of [RFC8305]) or performing local DNS64 functions need to discover the NAT64 prefix. ~~The n~~Network administrators SHOULD configure the first-hop routers to include PREF64 information in Router Advertisements ~~as per~~ ~~([RFC8781])~~ even if the network provides DNS64 (so hosts can use DNS64-based prefix discovery, [RFC7050]). [RFC9872] discusses this recommendation in more details.

4.3.4. Signalling DNS and DNS64 to Hosts

To provide IPv6-only endpoints with DNS recursive server addresses, the network administrator MUST configure the network to include the DNS servers IPv6 addresses into RA option, as per [RFC8106]. (While the DNS server addresses can also be provided via DHCPv6 [RFC9915], there is no guarantee that all IPv6-only endpoints have a DHCPv6 client enabled).

Commenté [MB41]: -- This MUST prevents operator deploy encrypted DNS if they want so (RFC9463)

~~Traditionally~~Some deployments use DNS64 (with NAT64) ~~is used~~ to enable IPv6-only endpoints to access IPv4-only destinations. However, using DNS64 has a number of drawbacks, such as:

- * DNSSEC Incompatibility: DNS64, when not enabled in the host, can interfere with DNSSEC validation.
- * Custom Resolvers: Endpoints or applications configured with custom resolvers or running recursive resolvers locally can not benefit from the DNS64 provided by the network. Such systems would need to rely on CLAT or local address synthesis instead (Section 7.4.9 discusses this scenario in more details).

- * Additional requirements for application: to benefit from DNS64, applications need to be IPv6-enabled, use DNS (do not use IPv4 literals). Many applications do not satisfy those requirements and, therefore, fail if the endpoint does not have an IPv4 address/native IPv4 connectivity. Existence of such applications is the main reason why transition to IPv6-only must be incremental (via IPv6-mostly phase) and requires the vast majority of endpoints to support CLAT.

Those concerns make DNS64 a suboptimal and undesirable solution long-term. To eliminate the needs for DNS64, the network shall signal PREF64 to endpoints (see Section 4.3.3), and the endpoints need to use the obtained PREF64 for performing local address synthesis and for CLAT.

It should be noted that not every application can benefit from local Address synthesis performed by the operating-systemOS, as it would require the application to use DNS (not IP address literals). Therefore, DNS64 is only required to support the following cases:

- * Systems and applications which perform local synthesis but do not support [RFC8781] for prefix discovery, and can only discover the NAT64 prefix using via DNS64 [RFC7050].
- * IPv6-only devices without CLAT (unless those endpoints are guaranteed never to need IPv4-only destinations, e.g., in case of a specialized network segment communicating solely with IPv6-capable destinations).

On the other hand, using DNS64 bypasses CLAT for IPv6-capable applications communicating with IPv4-only destinations, making the communication slightly more efficient. Additionally, it encourages dual-stack endpoints to utilize the IPv6 and NAT64 path to reach IPv4-only destinations. This operational pattern is useful for identifying IPv6-related issues and IPv4 dependencies at an earlier stage of the IPv6-mostly migration. For that reason, administrators might find it beneficial to run DNS64 even if it is not strictly necessary, provided the above mentioned drawbacks are addressed.

5. Solution Analysis

5.1. IPv6-only Compared to Dual-Stack

IPv6-mostly networks offer significant advantages over ~~traditional~~conventional dual-stack models where endpoints have both IPv4 and IPv6 addresses:

- * The deployment of dual-stack networks does not fundamentally resolve the core issue of IPv4 address exhaustion (both private and public IPv4 addresses). The IPv6-mostly approach, however, offers a mechanism to significantly reduce the IPv4 address space allocated to endpoints and reclaim existing IPv4 address space. Section 5.1.1 discusses this aspect in more details.

Commenté [MB42]: This depends on the requesting devices, and thus deployment specific.

* Controlled and incremental phase-out of IPv4: IPv6-mostly allows for the controlled phase out IPv4 from many endpoints, streamlining operations and improving overall network reliability at a measured and operator-controlled pace. While IPv6-mostly deployments do not inherently resolve the IPv4 address exhaustion issue (as previously noted), the approach effectively paves the road toward IPv6-only networks.

* Reduced Dependency on DHCPv4: As more devices operate seamlessly in IPv6-only mode, the criticality of DHCPv4 service diminishes significantly. This allows operators to scale down DHCPv4 infrastructure or, in some cases, even operate it with less stringent ~~service-Service_level_Level_objectives-Objectives~~ (SLOs), optimizing costs and resource allocation.

* Simplified troubleshooting due to reduced impact of Happy Eyeballs [RFC8305]: when both the client and the server are dual-stack, communication can happen either over IPv6 or over IPv4, and the protocol choice can change over time. This may ~~obscure-hide~~ network issues or make them intermittent, complicating the troubleshooting. In IPv6-mostly networks, IPv6-only hosts are much less affected by such an issue. Although ~~modern-some~~ versions of Happy

Eyeballs algorithm support falling back to IPv4 even in case of IPv6-only network, the delay before switching to IPv4 is so long that the fallback does not happen during regular operations. It should be noted, however, that when an IPv6-only ~~client~~ host communicates with an IPv4-only destination, the traffic may traverse CLAT or be sent as IPv6 towards the NAT64, depending on how the specific application is written.

Commenté [MB43]: Won't age well

Commenté [MB44]: What does this means?

The introduction of NAT64 within the IPv6-mostly model may appear as a drawback, as it inherits many of the limitations associated with NAT44, such as translating flows from different hosts to the same public IPv4 address, or scalability challenges caused by stateful operation [RFC6269]. However, ~~it's-it is~~ important to recognize that most IPv4-only or dual-stack networks already rely on NAT44 due to IPv4 address scarcity. For these networks, transitioning to an IPv6-mostly architecture would simply require enabling NAT64 ~~functionality~~ on existing NAT44 devices.

In a dual-stack network, flows originating from IPv6-~~only-only~~-capable endpoints to IPv4-only destinations are, in most cases, ~~translated by~~ a NAT44 ~~function~~. Within an IPv6-mostly environment, this role is fulfilled by NAT64. Consequently, the overall load on the NAT infrastructure remains effectively unchanged, with NAT64 essentially replacing NAT44 for a subset of traffic flows.

Commenté [MB45]: This can't be generalized unless we are more specific.

This is true for mobile/enterprise networks, but this is not true for fixed networks.

It is also worth emphasizing that NAT64 is employed solely for facilitating access to IPv4-~~only~~ resources. As the availability of IPv6-enabled resources continues to increase, the volume of traffic traversing NAT64 is expected to diminish proportionally.

As discussed in Section 7.3 coexistence of IPv6-only and IPv4-only

hosts on the same link might present challenges for on-link service discovery and on-link ~~peer2peer~~ communications. It might be ~~considered~~ a ~~disadvantage~~ of the IPv6-mostly model for networks where on-link communications between IPv4-only and IPv6-only devices is desirable.

5.1.1. Reducing IPv4 Address Consumption and Mitigating IPv4 Address Exhaustion

In IPv6-mostly networks, IPv4 addresses are assigned exclusively to endpoints that require them for operation. This approach reduces IPv4 address utilization within network segments compared to ~~traditional-conventional~~ dual-stack deployments. The proportion of devices operating in IPv6-only mode - and the resulting decrease in IPv4 consumption - is dependent on the specific endpoint footprint and may vary across different network environments. Furthermore, the number of devices requiring IPv4 is expected to decrease over time as more ~~operating-system~~ OSes implement support for CLAT, PREF64 discovery, and DHCPv4 option 108.

In IPv6-mostly deployments, the reduction ~~in-of~~ public IPv4 address consumption is only realized if the network assigns public IPv4 addresses directly to endpoints. For networks utilizing [RFC1918] addressing, the IPv6-mostly approach facilitates the conservation of private address space. While this benefit may be negligible for smaller networks, it is significant for large-scale enterprises and data centers that face private address exhaustion and seek to avoid the complexity of deploying multiple layers of NAT or mechanisms to handle overlapping addresses.

Additionally, while migrating from dual-stack to IPv6-mostly reduces IPv4 address consumption, existing networks require renumbering to downsize the IPv4 ~~netblocks-prefixes~~ assigned to those segments. For new deployments, however, administrators can leverage the benefits of IPv6-mostly without renumbering by provisioning smaller ~~netblocks-prefix blocks~~ from the outset.

5.2. IPv6-mostly Compared to a Dedicated IPv6-only Network

~~Traditional-Conventional~~ IPv6-only adoption involves separate networks alongside dual-stack ones. IPv6-mostly approach offers ~~significant~~ improvements, such as:

- * **Enhanced Scalability:** Separate IPv6-only networks double the number of SSIDs in wireless environments, causing channel congestion and degrading performance. IPv6-mostly doesn't require additional SSIDs. Similarly, it allows IPv4 and IPv6-only devices to coexist on the same wired VLANs, eliminating the need of additional Layer 2 segments, VLAN IDs, and their respective Layer 3 configurations.

Commenté [MB46]: ???

Commenté [MB47]: This also saves the complexity at the service layer as we don't need much complex infra to handle users with the same internal IPv4 address.

For example, this would translate for voice communication to have SBCs

* Operational Simplicity: Managing one network segment for all clients (regardless of IPv4 needs) simplifies operations, improves user experience (no more confusing SSID choices), and reduces support tickets related to mismatched connections. For wired connections, dynamic VLAN assignment becomes easier without device-specific IPv6 capability tracking.

* Optimized IPv4 address Consumption: User-selected dual-stack networks often lead to unnecessary IPv4 use, as users often connect IPv6-only capable devices to a dual-stack network. IPv6-mostly network allocates IPv4 addresses only when devices don't advertise IPv6-only capability (DHCPv4 Option 108).

* Improved Problem Visibility: User-selected fallback to dual-stack networks can mask issues with IPv6-only operation, hindering problem reporting and resolution. IPv6-mostly forces users to work through any issues, improving identification and enabling fixes for smoother long-term transition.

* Flexible, Incremental Transition: IPv6-mostly allows for gradual migration on a per-segment or even on per-device basis. Devices become IPv6-only only when deemed fully compatible with that mode.

Commenté [MB48]: Point to where this is discussed

6. Incremental Rollout Considerations

Migrating endpoints to IPv6-only fundamentally changes network dynamics by removing the IPv4 safety net. This includes the masking effect of ~~traditional~~ Happy Eyeballs algorithm [RFC8305]. IPv6 connectivity issues become far more prominent, including those previously hidden within dual-stack environments. Operators should be prepared to discover and troubleshoot issues in both endpoints and network infrastructure, even if the dual-stack network appeared problem-free.

Some rollout considerations are discussed in the following sections.

6.1. Opt-In and Opt-Out Modes

Before user-facing deployment, ~~the network~~ administrator SHOULD consider a dedicated IPv6-mostly proof-of-concept network for early adopters. While this temporarily sacrifices some IPv6-mostly benefits (Section 5.2→), it provides valuable operational experience and early issue detection. In a nutshell, the recommended approach contains two phases:

Commenté [MB49]: -- FUTs and pre-validations are common practice. Why do we need to have a SHOULD for such obvious thing?

I would avoid normative language here.

* Opt-In Phase: Invite tech-savvy early adopters to join an IPv6-mostly network and report issues. While response rates may be low, dedicated participants provide valuable troubleshooting data.

* Opt-Out Phase: Incrementally enable PREF64 signalling as well as Option 108. Allow selective disabling for problematic endpoints, in extreme case even by providing a separate network or rolling back IPv6-mostly in that particular subnet. Requiring users to file a problem report to opt-out provides a mechanism for gaining

visibility into user experience on IPv6-mostly network. This process facilitates the identification of related issues and expedites issue resolution.

In some scenarios (see Section 7.4.1) the administrator MAY keep a dual-stack network as a last resort fallback mechanism but SHOULD prevent users from connecting to it accidentally (e.g., it should be a hidden SSID with authentication enabled). For recurring temporary networks, for instance deployed on regular events such as a conference, operators may consider changing the last resort fallback network SSID between each event to help prevent inadvertent or long-term use of legacy resources, preventing devices which have the fallback SSID saved in their configuration from reconnecting automatically during the next event.

Commenté [MB50]: This one is a valid use of normative language

6.2. Per-Device and Per-Subnet Incremental Rollout

Limited control over endpoint configuration necessitates a per-subnet rollout, incrementally enabling first PREF64 signalling and then Option 108 processing in DHCPv4.

In most networks, all hosts on a given subnet receive the same RAs (unless mechanisms like per-client RA [RFC8273] are deployed), so signalling PREF64 via an RA option doesn't allow any per-endpoint opt out mechanism.

Some ~~operating systems~~ Oses provide a mechanism to enable DHCP Option 108 processing. In these cases, if the endpoints are managed, per-device rollout may be possible. Note that some Oses enable DHCP Option 108 support by default without providing configuration knobs to turn it off. Such systems become IPv6-only ~~the moment when~~ Option 108 is activated on the DHCPv4 server side.

Commenté [MB51]: Not expanded. Expand at first use

The following operational approach is RECOMMENDED:

- * PREF64 signalling: Enable PREF64 option for the router advertisement and/or DNS64 (see above). Especially if DNS64 is used, this might affect behaviour of some devices as the path via NAT64 would get generally preferred over native IPv4 path. Rollback at this stage affects the entire subnet.
- * DHCPv4 Server-Side Activation: Enable Option 108 processing. Some Oses automatically switch to IPv6-only. Rollback at this stage affects the entire subnet.
- * Controlled Endpoint Activation: Enable Option 108 on managed endpoints with per-device rollback possible.

6.3. Rollback Approach

For quick rollback, the network administrator SHOULD start with a minimal

Option 108 value (300 seconds, Section 3.4 of [RFC8925]) and possibly increase this value as the IPv6-mostly network proves reliable, reducing the likelihood of full-scale rollback. In most deployments, the load of DHCPv4 infrastructure caused by each endpoint restarting the DHCP handshake every 300 seconds would be negligible. However,

for large networks containing a high number of IPv6-only capable clients, the default value of 300 seconds may be too low. Even if each IPv6-only capable client sends only one DHCPDISCOVER every 300 seconds, the collective traffic can still create a substantial load on the DHCPv4 infrastructure. Administrators are advised to monitor this load and adjust the Option 108 value as necessary.

7. Operational Considerations

7.1. Address Assignment Policy

As outlined in Section 6.3 of [RFC6877], CLAT requires either a dedicated IPv6 prefix or, if unavailable, a dedicated IPv6 address. At the time of this writing, all implementations use SLAAC for CLAT address acquisition. Therefore, to enable CLAT functionality within IPv6-mostly network segments, first-hop routers MUST be configured to advertise a Prefix Information Option (PIO, [RFC4861]) containing a globally routable SLAAC-suitable prefix with the 'Autonomous Address-Configuration' (A) flag set to one.

7.2. Extension Headers

~~Being an IPv6-specific concept,~~ IPv6 extension headers are often neglected or even ~~explicitly~~explicitly prohibited by security policies in dual-stack networks. The issues caused by blocking extension headers might be masked by the ~~presence~~presence of Happy Eyeballs but become highly visible when there is no IPv4 to fallback to.

The network security policy SHOULD permit at least the following extension headers:

* Fragment Header (Section 4.5 of [RFC8200]). Section 7.4.5 discusses the fragmentation in more details.

* ESP Header, which is used for IPsec traffic, such as VPN and Wi-Fi Calling, unless establishing IPsec tunnels is not desirable.

~~Network and security administrators~~ should consider recommendations from Section 2.1 of [RFC7045], as well as [RFC9673] when developing the security policies for other IPv6 extension headers.

7.3. On-link Communication and Service Discovery

~~Shared links is-are often used for automatic~~ discovery of neighboring devices and their services by means of different protocols like Multicast DNS [RFC6762]. Many devices and/or services are built on an assumption that devices on the same link also share the same set of address families or at least they have one common address family shared between all of them.

In IPv6-mostly, this assumption is not valid as there might be both IPv4-only and IPv6-only devices on the same link. As per Section 20 of [RFC6762], this means that the link has two unrelated ".local." zones, one for each address family. Discovery of devices across different address families is impossible, unless some sort of relay is deployed on the link.

Commenté [MB52]: Explain in the text what this one is not a MUST

Commenté [MB53]: Who do we distinguish both? Isn't «network administrators» generic enough for the purpose of this doc?

Commenté [MB54]: -- There is not such notion in 6762

This problem, however, is limited to networks where client-to-client on-link communications are desired and permitted by security policies. For networks where client isolation is enabled (~~it's-it is~~ often a case for enterprise networks and public ~~Wi-FiWi-Fi~~), this issue is not a concern. If ~~on-link peer2peer communication~~ is required, the operator needs to ensure all participating devices are either dual-stack or IPv6-mostly.

Commenté [MB55]: ??

7.4. Typical Issues

IPv6-mostly networks expose hidden issues by removing the IPv4 safety net. While implementation bugs vary greatly and are beyond the scope of this document, this section ~~focuses focus~~ on common problems caused by configuration, topology, or design choices. It is important to emphasize that these issues frequently pre-exist in dual-stack environments but remain undetected due to IPv4 fallback mechanisms.

The problems described herein may not be applicable to every IPv6-mostly deployment. However, operators are advised to verify that these issues are not present within their infrastructure prior to initiating an IPv6-mostly ~~migration-or~~ deployment.

7.4.1. ~~Hosts-Endpoints~~ with Disabled or Disfunctional IPv6

Historically, tech support often advised disabling IPv6 as a quick workaround, leading to devices with disabled IPv6. Similarly, corporate IT may have disabled or filtered IPv6 under the assumption that ~~it's-it is~~ not widely used. Such endpoints requesting Option 108 will fail to connect in an IPv6-mostly network, as they won't receive IPv4 addresses and IPv6 is disabled.

Administrators controlling endpoints SHOULD ensure those endpoints have IPv6 enabled and operational before transitioning the network to IPv6-mostly mode. This includes verifying protocol enablement as well as validation of any central or discreetly managed host based firewalls that could potentially interfere with proper IPv6 function that may be masked by the presence of IPv4, and therefore exposed with its removal. ~~Absent such checks, the connectivity may not be provided as intended.~~

Commenté [MB56]: Add some text to explain what happens if SHOULD is not followed.

7.4.2. Network Extension

~~IPv4's NAT44~~ allows endpoints to extend connectivity to downstream systems without upstream network awareness or permission. Note that "endpoint" describes a role rather than a fixed device category: some endpoints also extend network connectivity to other physical or logical systems, thereby additionally acting as routers. Examples include personal computers running virtual machines or containers that route traffic to them, and mobile phones with tethering enabled that act as a router for tethered devices, potentially without the operator's knowledge or consent. This creates challenges in IPv6-mostly deployments where endpoints lack IPv4 addresses:

Commenté [MB57]: I don't parse this

Solutions and trade-offs:

- * Using DHCPv6-PD to allocate prefixes to endpoints ~~+[RFC9663+-]~~: Provides downstream systems with IPv6 addresses and native connectivity.
- * Enabling the CLAT function on the endpoint:- This scenario is similar to the Wireline Network Architecture described in Section 4.1 of [RFC6877]. The downstream systems would receive IPv4 addresses and their IPv4 traffic would be translated to IPv6 by the endpoint. ~~However, However~~-this approach leads to the downstream systems using IPv4 only and not benefiting from end-to-end IPv6 connectivity. To enable IPv6 benefits, combine this with IPv6 Prefix Delegation as ~~discussed-listed~~ above.
- * Bridging and Neighbor Discovery (ND) ND-Proxy: The endpoint bridges IPv6 traffic and masks downstream devices behind its MAC address. This can lead to scalability issues (Section 7.4.3) due to the single MAC being mapped to many IPv6 addresses.

7.4.3. Multiple Addresses per Device

Unlike IPv4, where endpoints typically have a single IPv4 address per interface, IPv6 endpoints inherently use multiple addresses such as:

- * Link-local address
- * Temporary address (default on endpoints for privacy)
- * Stable address (for long-term identification)
- * CLAT address (in IPv6-mostly/IPv6-only networks)

Endpoints with containers, namespaces, or ND Proxyproxy functions may have

even more addresses. This poses challenges for network infrastructure devices (SAVI switches, wireless access points, etc.) that map MAC addresses to IPv6 addresses, often with limits to prevent resource exhaustion or denial-of-service (DoS) attacks. When the number of IP addresses per MAC limit is exceeded, infrastructure devices behavior varies across implementations, leading to inconsistent connectivity loss and other unexpected behavior: while some systems drop new addresses, others delete older entries, causing previously functional addresses to lose connectivity. In all those cases endpoints and applications don't receive explicit signalling about the address becoming unusable.

This problem, while impacting dual-stack endpoints as well, is much less visible for dual-stack devices:

- * On a dual-stack endpoint Happy Eyeballs might obscure the issue by falling back to IPv4.
- * Using dedicated IPv6 addresses for CLAT instances increases the number of IPv6 addresses required on IPv6-only nodes compared to

Commenté [MB58]: I don't understand this.

Commenté [MB59]: Can we cite a reference?

Commenté [MB60]: Please expand

dual-stack ones.

- * Some traffic flows (e.g., belonging to IPv4-only applications) would be using IPv4 on a dual-stack endpoint but go through CLAT on an IPv6-only one. Such flows would be unaffected on a dual-stack network but experience an outage if the CLAT address is blocked by the network.

Allocating prefixes to endpoints via DHCPv6-PD ~~+[RFC9663]~~ allows to eliminate the issue and to address the corresponding scalability concerns. Operators of large-scale IPv6-mostly networks might benefit from using DHCPv6 prefix delegation for IPv6-only and IPv6-enabled endpoints. In that case, ~~the network~~ administrators also need to ensure that the first-hop routers set the P-flag [RFC9762] in RAs. However, the deployment model described in [RFC9663] might not ~~yet~~ be supported by all endpoints and network infrastructure devices. ~~The network~~ Network administrators SHOULD ensure that the deployed network infrastructure devices allow sufficient number of IPv6 addresses to be mapped to a client's MAC and SHOULD monitor for events, indicating that the limit has been reached (such as syslog messages ~~[RFC5424]~~, etc-).

7.4.4. Host Mobility and Renumbering

Networks employing dynamic VLAN assignment (e.g., based on 802.1x or MAC-based authentication) can cause endpoints to move between VLANs and IPv6 subnets. As client operating systems do not always handle changes in link-layer state (e.g., VLAN changes) correctly, this mobility often leads to inconsistent IP stack behavior on operating systems, resulting in the persistence of old subnet addresses and potential connectivity issues due to incorrect source address selection. In such networks, the administrator SHOULD configure the link-local addresses of the first-hop routers (such as link-local addresses assigned to router interfaces or the corresponding ~~VRRPv3~~ Virtual Router Redundancy Protocol (~~VRRPv3~~) [RFC9568] virtual link-local IP address) to be unique within the mobility domain (the set of links the host can move between).

For example, if there are two VLANs using subnets 2001:db8:1:a::/64 and 2001:db8:1:b::/64, the administrator might configure the virtual router link-local addresses as 'fe80::2001:db8:1:a' and 'fe80::2001:db8:1:b', respectively.

[I-D.link-6man-gulla] provides further analysis and discusses this approach in more detail.

7.4.5. Fragmentation

As the basic IPv6 header is 20 bytes longer than the IPv4 header, translating from IPv4 to IPv6 can result in packets exceeding the path MTU on the IPv6 side. In ~~that-such a~~ case, NAT64 creates IPv6 packets with the Fragment Header (see Section 4 of [RFC7915] for more details). As per [RFC7915], ~~by~~ by default the translator fragments IPv4 packets so that they fit in 1280-byte IPv6 packets. It means that all IPv4 packets larger than 1260 bytes are fragmented (or

Commenté [MB61]: This was expired. Do we need to have this in the final RFC?

dropped if the DF bit is set). While the translator is expected to send an ICMPv4 "Fragmentation Needed" error message back to the host when dropping such packets (see Section 4.1 of [RFC7915]), generating ICMP messages is often rate-limited and, therefore, can not be considered reliable.

Administrators SHOULD maximize the path MTU on the IPv6 side (from the translator to IPv6-only hosts) to minimize fragmentation. NAT64 devices SHOULD be configured to use the actual path MTU on the IPv6 side when fragmenting IPv4 packets.

Another common case of IPv6 fragmentation is the use of protocols like DNS and RADIUS, where the server response needs to be sent as a single UDP datagram. Network security policies MUST allow IPv6 fragments for permitted UDP traffic (e.g., DNS ~~or~~ RADIUS) where datagram responses are required. Allowing IPv6 fragments for permitted TCP traffic is RECOMMENDED unless the network infrastructure reliably performs TCP MSS clamping.

To ensure reliable Path MTU Discovery (PMTUD, [RFC8201]), Section 4.3.1 of [RFC4890] requires that network security policies MUST permit the passage of ICMPv6 Type 2 (Packet Too Big, [RFC4443]) messages to and from all IPv6-enabled devices. This recommendation is fundamentally no different from the established practice in IPv4 networks, where ICMP Type 3 Code 4 (Fragmentation Needed and DF set) messages are essential for PMTUD.

7.4.5.1. Atomic Fragments

Section 4 of [RFC6145] specified that an IPv4 packet without the Don't Fragment (DF) bit set, when translated to IPv6, should include an IPv6 Fragment Header. Such packets are known as atomic fragments. However, [RFC7915] obsoleted [RFC6145] and deprecated the algorithm that generates these IPv6 atomic fragments. It has been observed that certain [RFC7915] non-compliant **PLAT** implementations continue to generate IPv6 atomic fragments. This behavior causes unintended connectivity issues, particularly on endpoints where firewalls are enabled. Administrators should verify that the PLAT devices do not generate atomic fragments by default. If atomic fragments are generated, the administrator should configure the PLAT devices to disable atomic fragments generation, provided a configuration mechanism is available.

7.4.6. IPv4 UDP Packets with Zero Checksum

Translating IPv4 UDP packet with zero checksum to IPv6 requires the translator to calculate the checksum and therefore has performance implication. As per Section 4.5 of [RFC7915], translators are allowed to drop IPv4 UDP packet with zero checksum. Additionally, Section 1.2 of [RFC7915] says: "Fragmented IPv4 UDP packets that do not contain a UDP checksum (i.e., the UDP checksum field is zero) are not of significant use on the Internet, and in general will not be translated by the IP/ICMP translator~~....~~". However, when the translator is configured to forward the packet without a UDP checksum, the fragmented IPv4 UDP packets will be translated."

Commenté [MB62]: -- These do not usually cross a NAT64. Why these are listed here?

Commenté [MB63]: NAT64?

a mis en forme : Français (France)

a mis en forme : Français (France)

It has been observed that some systems (such as 3GPP-~~ePDG~~, Evolved Packet Data Gateway (ePDG)) might generate IPv4 UDP packets with zero checksum. Those packets could be dropped by the PLAT devices, impacting WiFi calling on IPv6-only endpoints connecting to those ePDGs. As Section 4.5 of [RFC7915] recommends that the translators have a configuration knob to calculate an IPv6 checksum and forward the packet instead of dropping it, the administrator should consider enabling that option if such an issue is encountered.

7.4.7. Representing IPv6 Addresses by CLAT

Certain CLAT implementations face challenges when translating incoming IPv6 packets with native (non-synthesized) source addresses (e.g., ICMPv6 packets sent by intermediate hops on the path). This lack of standardized translation mechanisms can lead to:

- * Incomplete Traceroute: Omission of IPv6-only hops between the endpoint and NAT64 translator, hindering troubleshooting.
- * Path MTU Discovery Issues: Potential disruptions in the PMTU discovery process.

[I-D.ietf-v6ops-icmpext-xlat-v6only-source] ~~proposes~~ defines a solution for signalling the actual non-synthesized IPv6 source address while translating ICMPv6 error messages.

7.4.8. IPv4-Dependencies in Network Admission Control

Certain layer 2 network devices (e.g., wireless access points, controllers) may enforce a policy where a host's network access is contingent upon successful DHCP IPv4 address assignment or tied to results of DHCP snooping. Consequently, hosts supporting Option 108 may experience network access denial or disconnection, as they are not assigned IPv4 addresses. To facilitate a smooth migration to IPv6-mostly, operators are advised to update their network access policies to eliminate any hard dependencies on the assignment of IPv4 addresses to hosts.

7.4.9. Custom DNS Configuration on Endpoints

In IPv6-mostly networks without PREF64 in RAs, hosts rely on DNS64 [RFC7050] ~~to discover the NAT64 prefix for CLAT operation.~~

[RFC8880] requires that queries for AAAA resource records of "ipv4only.arpa." ~~"MUST be sent to the recursive resolvers provided by the network, MUST be sent to the DNS recursive resolver address(es) the client learned via the configuration machinery for that specific client interface"~~ not to the resolvers configured manually, local recursive resolvers ~~etc.~~ ~~However, However~~ some implementations do not comply

with [RFC8880] and, when configured with custom DNS resolvers (e.g., public or corporate DNS), may bypass the network-provided DNS64, preventing NAT64 prefix discovery and hindering CLAT functionality.

Where feasible, network administrators SHOULD include PREF64 in RAs within

a mis en forme : Anglais (États-Unis)

Commenté [MB64]: Use the exact text from 8880

IPv6-mostly networks to minimize reliance on DNS64. Administrators need to be aware of the potential for CLAT failures when endpoints use custom resolvers in environments lacking PREF64.

Similarly, some VPN clients are known to alter DNS resolution behavior on the endpoint - for example by inserting their own resolver ahead of the network-provided one, or by suppressing AAAA responses even when native IPv6 connectivity remains available - preventing those devices from discovering the NAT64 prefix via the [RFC7050] mechanism. If the network doesn't provide the NAT64 prefix in RAs or the endpoint doesn't support [RFC8781], the endpoint might not be able to discover the NAT64 prefix and therefore would fail to enable CLAT.

7.4.10. Virtual Private Network (VPN) Policy

VPN clients and policies have been observed to interfere with IPv6 connectivity for endpoints in an IPv6-mostly network through several distinct mechanisms. Administrators and operators should be aware of each individually, since detection and remediation differ across the problem space. The following list is not exhaustive, but represents the most commonly observed behaviors:

- * Disabling the IPv6 protocol stack on the host entirely, as part of the VPN profile, as described above.
- * Suppressing AAAA responses or otherwise causing name resolution to return only IPv4 addresses while the IPv6 interface and underlying connectivity remain fully functional. This can be notably harder to detect than a completely disabled IPv6 address family since IPv6 appears operational at the interface level.
- * Installing a blackhole or null route for the IPv6 default route while leaving the IPv6 stack enabled.
- * Applying host-based or network firewall rules that block IPv6 traffic while the VPN is active. This can be difficult to both detect and remediate as changes to local security tooling may be administratively restricted.

The specific behavior depends heavily on how the VPN client integrates with the host's network stack: some VPNs present a logical interface and rely on the host's own routing table to select tunneled versus local traffic, while others intercept traffic at a lower layer and enforce policy directly. This document does not attempt to enumerate behavior across specific VPN products or operating systems; administrators should verify actual behavior in their own environment rather than assume IPv6 is either fully passed or fully blocked.

See Section 7.4.9 for a related discussion of DNS-level interference with NAT64 prefix discovery.

8. Security Considerations

The ~~proposed-IPv6-mostly~~ deployment scenario inherits security considerations of

IPv6 (see [RFC9099]). As IPv6-only ~~device-devices~~ can not fall back to IPv4 if IPv6 connectivity is not available or impacted by a malicious actor, any attack affecting IPv6 connectivity would have much more drastic outcome, comparing to dual-stack networks.

By signalling a rogue NAT64 prefix to a host, a malicious actor can:

- * cause a DoS attack, if the network does not provide NAT64 functions (for that prefix or at all);
- * implement MitM-on-path attack by intercepting traffic to the rogue prefix.

To countermeasure this attack vector, the network administrators SHOULD configure the first-hop routers to include PREF64 information in Router Advertisements as per [RFC8781] and ensure that Layer 2 security measures such as RA-Guard ([RFC6105]) are in place. Section 7 of [RFC8781] discusses this topic in more details. Additionally, [I-D.ietf-v6ops-clat] discusses security implications of enabling CLAT if native IPv4 connectivity is available and recommends disabling CLAT in that case.

Security considerations of using DHCPv4 Option 108 are documented in Section 6 of [RFC8925].

9. Privacy Considerations

This document does not introduce any new privacy considerations. IPv6-mostly networks have the same privacy considerations as other Dual-stacked or IPv6-only networks.

10. IANA Considerations

This memo does not introduce any requests to IANA.

11. References

11.1. Normative References

- [RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997, <<https://www.rfc-editor.org/info/rfc2119>>.
- [RFC6105] Levy-Abegnoli, E., Van de Velde, G., Popoviciu, C., and J. Mohacsi, "IPv6 Router Advertisement Guard", RFC 6105, DOI 10.17487/RFC6105, February 2011, <<https://www.rfc-editor.org/info/rfc6105>>.
- [RFC6146] Bagnulo, M., Matthews, P., and I. van Beijnum, "Stateful NAT64: Network Address and Protocol Translation from IPv6 Clients to IPv4 Servers", RFC 6146, DOI 10.17487/RFC6146, April 2011, <<https://www.rfc-editor.org/info/rfc6146>>.
- [RFC6877] Mawatari, M., Kawashima, M., and C. Byrne, "464XLAT: Combination of Stateful and Stateless Translation", RFC 6877, DOI 10.17487/RFC6877, April 2013,

<<https://www.rfc-editor.org/info/rfc6877>>.

- [RFC7050] Savolainen, T., Korhonen, J., and D. Wing, "Discovery of the IPv6 Prefix Used for IPv6 Address Synthesis", RFC 7050, DOI 10.17487/RFC7050, November 2013, <<https://www.rfc-editor.org/info/rfc7050>>.
- [RFC8200] Deering, S. and R. Hinden, "Internet Protocol, Version 6 (IPv6) Specification", STD 86, RFC 8200, DOI 10.17487/RFC8200, July 2017, <<https://www.rfc-editor.org/info/rfc8200>>.
- [RFC8106] Jeong, J., Park, S., Beloeil, L., and S. Madanapalli, "IPv6 Router Advertisement Options for DNS Configuration", RFC 8106, DOI 10.17487/RFC8106, March 2017, <<https://www.rfc-editor.org/info/rfc8106>>.
- [RFC8174] Leiba, B., "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words", BCP 14, RFC 8174, DOI 10.17487/RFC8174, May 2017, <<https://www.rfc-editor.org/info/rfc8174>>.
- [RFC8781] Colitti, L. and J. Linkova, "Discovering PREF64 in Router Advertisements", RFC 8781, DOI 10.17487/RFC8781, April 2020, <<https://www.rfc-editor.org/info/rfc8781>>.
- [RFC8925] Colitti, L., Linkova, J., Richardson, M., and T. Mrugalski, "IPv6-Only Preferred Option for DHCPv4", RFC 8925, DOI 10.17487/RFC8925, October 2020, <<https://www.rfc-editor.org/info/rfc8925>>.

11.2. Informative References

- [RFC1918] Rekhter, Y., Moskowitz, B., Karrenberg, D., de Groot, G. J., and E. Lear, "Address Allocation for Private Internets", BCP 5, RFC 1918, DOI 10.17487/RFC1918, February 1996, <<https://www.rfc-editor.org/info/rfc1918>>.
- [RFC4443] Conta, A., Deering, S., and M. Gupta, Ed., "Internet Control Message Protocol (ICMPv6) for the Internet Protocol Version 6 (IPv6) Specification", STD 89, RFC 4443, DOI 10.17487/RFC4443, March 2006, <<https://www.rfc-editor.org/info/rfc4443>>.
- [RFC4787] Audet, F., Ed. and C. Jennings, "Network Address Translation (NAT) Behavioral Requirements for Unicast UDP", BCP 127, RFC 4787, DOI 10.17487/RFC4787, January 2007, <<https://www.rfc-editor.org/info/rfc4787>>.
- [RFC4861] Narten, T., Nordmark, E., Simpson, W., and H. Soliman, "Neighbor Discovery for IP version 6 (IPv6)", RFC 4861, DOI 10.17487/RFC4861, September 2007, <<https://www.rfc-editor.org/info/rfc4861>>.
- [RFC4890] Davies, E. and J. Mohacsi, "Recommendations for Filtering ICMPv6 Messages in Firewalls", RFC 4890, DOI 10.17487/RFC4890, May 2007, <<https://www.rfc-editor.org/info/rfc4890>>.

Commenté [MB65]: This is normative for:

To ensure reliable Path MTU Discovery (PMTUD, [RFC8201]),
Section 4.3.1 of [RFC4890] requires that network security policies
MUST permit the passage of ICMPv6 Type 2 (Packet Too Big, [RFC4443])

Commenté [MB66]: -- this is normative for

Therefore, to enable CLAT functionality within IPv6-mostly network segments, first-hop routers MUST be configured to advertise a Prefix Information Option (PIO, [RFC4861]) containing a

[RFC6052] Bao, C., Huitema, C., Bagnulo, M., Boucadair, M., and X. Li, "IPv6 Addressing of IPv4/IPv6 Translators", RFC 6052, DOI 10.17487/RFC6052, October 2010, <<https://www.rfc-editor.org/info/rfc6052>>.

Commenté [MB67]: I think this one is normative to digest the various prefixes, address synthesis

[RFC6145] Li, X., Bao, C., and F. Baker, "IP/ICMP Translation Algorithm", RFC 6145, DOI 10.17487/RFC6145, April 2011, <<https://www.rfc-editor.org/info/rfc6145>>.

[RFC6147] Bagnulo, M., Sullivan, A., Matthews, P., and I. van Beijnum, "DNS64: DNS Extensions for Network Address Translation from IPv6 Clients to IPv4 Servers", RFC 6147, DOI 10.17487/RFC6147, April 2011, <<https://www.rfc-editor.org/info/rfc6147>>.

[RFC6535] Huang, B., Deng, H., and T. Savolainen, "Dual-Stack Hosts Using "Bump-in-the-Host" (BIH)", RFC 6535, DOI 10.17487/RFC6535, February 2012, <<https://www.rfc-editor.org/info/rfc6535>>.

[RFC6598] Weil, J., Kuarsingh, V., Donley, C., Liljenstolpe, C., and M. Azinger, "IANA-Reserved IPv4 Prefix for Shared Address Space", BCP 153, RFC 6598, DOI 10.17487/RFC6598, April 2012, <<https://www.rfc-editor.org/info/rfc6598>>.

[RFC6762] Cheshire, S. and M. Krochmal, "Multicast DNS", RFC 6762, DOI 10.17487/RFC6762, February 2013, <<https://www.rfc-editor.org/info/rfc6762>>.

[RFC6724] Thaler, D., Ed., Draves, R., Matsumoto, A., and T. Chown, "Default Address Selection for Internet Protocol Version 6 (IPv6)", RFC 6724, DOI 10.17487/RFC6724, September 2012, <<https://www.rfc-editor.org/info/rfc6724>>.

[RFC6888] Perreault, S., Ed., Yamagata, I., Miyakawa, S., Nakagawa, A., and H. Ashida, "Common Requirements for Carrier-Grade NATs (CGNs)", BCP 127, RFC 6888, DOI 10.17487/RFC6888, April 2013, <<https://www.rfc-editor.org/info/rfc6888>>.

[RFC7045] Carpenter, B. and S. Jiang, "Transmission and Processing of IPv6 Extension Headers", RFC 7045, DOI 10.17487/RFC7045, December 2013, <<https://www.rfc-editor.org/info/rfc7045>>.

[RFC7755] Anderson, T., "SIIT-DC: Stateless IP/ICMP Translation for IPv6 Data Center Environments", RFC 7755, DOI 10.17487/RFC7755, February 2016, <<https://www.rfc-editor.org/info/rfc7755>>.

[RFC7915] Bao, C., Li, X., Baker, F., Anderson, T., and F. Gont, "IP/ICMP Translation Algorithm", RFC 7915, DOI 10.17487/RFC7915, June 2016, <<https://www.rfc-editor.org/info/rfc7915>>.

[RFC8201] McCann, J., Deering, S., Mogul, J., and R. Hinden, Ed., "Path MTU Discovery for IP version 6", STD 87, RFC 8201, DOI 10.17487/RFC8201, July 2017, <<https://www.rfc-editor.org/info/rfc8201>>.

- [RFC8215] Anderson, T., "Local-Use IPv4/IPv6 Translation Prefix", RFC 8215, DOI 10.17487/RFC8215, August 2017, <<https://www.rfc-editor.org/info/rfc8215>>.
- [RFC8273] Brzozowski, J. and G. Van de Velde, "Unique IPv6 Prefix per Host", RFC 8273, DOI 10.17487/RFC8273, December 2017, <<https://www.rfc-editor.org/info/rfc8273>>.
- [RFC8305] Schinazi, D. and T. Pauly, "Happy Eyeballs Version 2: Better Connectivity Using Concurrency", RFC 8305, DOI 10.17487/RFC8305, December 2017, <<https://www.rfc-editor.org/info/rfc8305>>.
- [RFC8880] Cheshire, S. and D. Schinazi, "Special Use Domain Name 'ipv4only.arpa'", RFC 8880, DOI 10.17487/RFC8880, August 2020, <<https://www.rfc-editor.org/info/rfc8880>>.
- [RFC9099] Vyncke, É., Chittimaneni, K., Kaeo, M., and E. Rey, "Operational Security Considerations for IPv6 Networks", RFC 9099, DOI 10.17487/RFC9099, August 2021, <<https://www.rfc-editor.org/info/rfc9099>>.
- [RFC9568] Lindem, A. and A. Dogra, "Virtual Router Redundancy Protocol (VRRP) Version 3 for IPv4 and IPv6", RFC 9568, DOI 10.17487/RFC9568, April 2024, <<https://www.rfc-editor.org/info/rfc9568>>.
- [RFC9663] Colitti, L., Linkova, J., Ed., and X. Ma, Ed., "Using DHCPv6 Prefix Delegation (DHCPv6-PD) to Allocate Unique IPv6 Prefixes per Client in Large Broadcast Networks", RFC 9663, DOI 10.17487/RFC9663, October 2024, <<https://www.rfc-editor.org/info/rfc9663>>.
- [RFC9673] Hinden, R. and G. Fairhurst, "IPv6 Hop-by-Hop Options Processing Procedures", RFC 9673, DOI 10.17487/RFC9673, October 2024, <<https://www.rfc-editor.org/info/rfc9673>>.
- [RFC9762] Colitti, L., Linkova, J., Ma, X., Ed., and D. Lamparter, "Using Router Advertisements to Signal the Availability of DHCPv6 Prefix Delegation to Clients", RFC 9762, DOI 10.17487/RFC9762, June 2025, <<https://www.rfc-editor.org/info/rfc9762>>.
- [RFC9872] Buraglio, N., Jensen, T., and J. Linkova, "Recommendations for Discovering IPv6 Prefix Used for IPv6 Address Synthesis", RFC 9872, DOI 10.17487/RFC9872, September 2025, <<https://www.rfc-editor.org/info/rfc9872>>.
- [RFC9915] Mrugalski, T., Volz, B., Richardson, M., Jiang, S., and T. Winters, "Dynamic Host Configuration Protocol for IPv6 (DHCPv6)", STD 102, RFC 9915, DOI 10.17487/RFC9915, January 2026, <<https://www.rfc-editor.org/info/rfc9915>>.
- [I-D.ietf-v6ops-clatlon]
Colitti, L., Linkova, J., and T. Jensen, "464XLAT Customer-side Translator (CLAT): Node Behavior and Recommendations", Work in Progress, Internet-Draft, draft-

ietf-v6ops-claton-16, 5 March 2026,
<<https://datatracker.ietf.org/doc/html/draft-ietf-v6ops-claton-16>>.

[I-D.link-6man-gulla]

Linkova, J., "Using Prefix-Specific Link-Local Addresses to Improve SLAAC Robustness", Work in Progress, Internet-Draft, draft-link-6man-gulla-01, 3 March 2025,
<<https://datatracker.ietf.org/doc/html/draft-link-6man-gulla-01>>.

[I-D.ietf-v6ops-icmpeext-xlat-v6only-source]

Lamparter, D. E. and J. Linkova, "Using Dummy IPv4 Address and Node Identification Extensions for IP/ICMP translators (XLAATs)", Work in Progress, Internet-Draft, draft-ietf-v6ops-icmpeext-xlat-v6only-source-01, 6 November 2025,
<<https://datatracker.ietf.org/doc/html/draft-ietf-v6ops-icmpeext-xlat-v6only-source-01>>.

[I-D.ietf-v6ops-nat64-wkp-1918]

Kumari, W. and J. Linkova, "Using the Well-Known IPv6 Prefix to Represent Non-Global IPv4 Addresses", Work in Progress, Internet-Draft, draft-ietf-v6ops-nat64-wkp-1918-08, 26 August 2026,
<<https://datatracker.ietf.org/doc/html/draft-ietf-v6ops-nat64-wkp-1918-08>>.

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Authors' Addresses

Nick Buraglio
Energy Sciences Network
IL
United States of America
Email: buraglio@forwardingplane.net, buraglio@es.net

Ondrej Caletka
RIPE NCC
Stationsplein 11
Amsterdam
Netherlands
Email: ondrej.caletka@ripe.net

Jen Linkova
Google
1 Darling Island Rd
Pyrmont NSW 2009
Australia

Email: furry13@gmail.com